

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
url = "https://raw.githubusercontent.com/dD2405/Twitter_Sentiment_Analysis/master/train.csv"

df = pd.read_csv(url)

df.head()
```

	id	label	tweet
0	1	0	@user when a father is dysfunctional and is s...
1	2	0	@user @user thanks for #lyft credit i can't us...
2	3	0	bihday your majesty
3	4	0	#model i love u take with u all the time in ...
4	5	0	factsguide: society now #motivation

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.columns
df.shape
df.head()
```

	id	label	tweet
0	1	0	@user when a father is dysfunctional and is s...
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```
df['label'].value_counts()
```

	count
label	
0	29720
1	2242

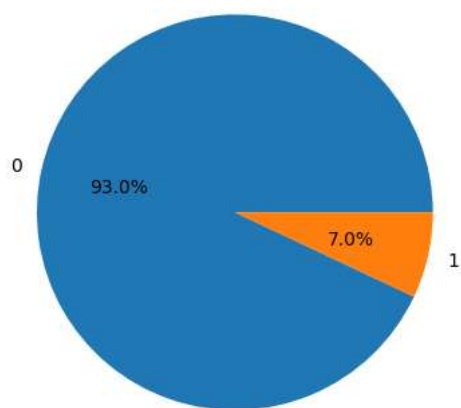
```
dtype: int64
```

```
import matplotlib.pyplot as plt

df['label'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Sentiment Distribution")
plt.ylabel("")
plt.show()
```

Sentiment Distribution



```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
```

```
X = df['tweet']
y = df['label']
```

```
vectorizer = CountVectorizer()
```

```
X = vectorizer.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = MultinomialNB()
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.9425934615986234
```

```
sample_tweet = ["I am very happy today"]
```

```
sample_vector = vectorizer.transform(sample_tweet)
```

```
prediction = model.predict(sample_vector)
```

```
if prediction[0] == 0:
    print("Positive Sentiment 😊")
else:
    print("Negative Sentiment 😞")
```

```
Positive Sentiment 😊
```

```
tweets = [
    "I love this product",
    "This service is terrible",
    "Today is a normal day"
]

for tweet in tweets:
    pred = model.predict(vectorizer.transform([tweet]))
    sentiment = "Positive 😊" if pred[0] == 0 else "Negative 😞"
    print(tweet, "->", sentiment)
```

```
I love this product -> Positive 😊
This service is terrible -> Positive 😊
Today is a normal day -> Positive 😊
```

